

Watch's the future like?

Watches are no longer looked at as just time pieces, and probably every single watch manufacturer is planning on getting the unthinkable into a watch. Already, sports-watches have a variety of features—temperature sensors and altimeters, slide rules and tools for divers.

The Fossil Wrist Net and Suunto N3 are examples of what watches can become. Fossil has also worked towards making watch-PDAs, although they're expensive.

These are basically decent PDA-watch



Calling secret agent Xact, here's your next mission!

hybrids, with standard Palm PDA features—essentially, a personal organiser disguised as a watch. Cell phone-watch form factors have

also been tried out. Last year, for instance, NTT DoCoMo was looking at releasing the WRISTOMO, a watch that lets you make phone calls, send and receive e-mail and exchange data at 64 Kbps.

The good old two-way communicator may well be coming alive, with Xact Communications' Wristlinx two-way radio, which puts a radio transceiver into the watch. It seems that next-gen watches will either be personal communication devices or personal organiser devices.

esting; SCA Data Systems and Microsoft worked on creating a noise tolerant, high bit-rate solution that uses the FM sub-carrier technology. The system currently has a throughput rate of 12 Kbps that is sufficient for data to flow continuously into the watch. Microsoft tells us the watches run on an embedded software platform that incorporates several .NET tools and has been optimised for miniaturisation and minimal power consumption.

Tuning in

Microsoft calls it DirectBand. The name collectively applies to the radio receiving units and the wide area network created over existing FM radio networks. Direct-Band works in association with US radio broadcasters, using unused frequencies in the radio spectrum for content broadcasts.

While signing up for the MSN Direct service, customers enter a unique personalised reception code—the 256-bit identification register etched by lasers into the Arm7 core—and set up the kinds of personalised content feeds they want.

Currently, you can choose from public feeds such as news, weather, sports and stock tickers as well as private feeds such as Outlook alerts and instant messages, all of which, Microsoft says, are encrypted. A plug-in connects Outlook to the MSN Direct service. Then tasks and meeting alerts can be set to make the SPOT watch buzz you. To send instant messages via MSN Messenger, right-click on the user-name and choose to send messages.

All the information requested is routed in through MSN Direct's data centre and then transmitted out on the FM airwaves via local radio stations, en masse. The RF chip picks up all the radio content being streamed out, and sifts through all of it to pick up and display only the content specific to its unique personalisation code.

How good is it?

The service sports several cool features; one among them is the ability to get fun faces for each of the watches. Yet another is the roaming feature. If you happen to move out of one coverage zone and into another, you don't need to reset the time. MSN Direct automatically does that for you, so you always have the correct local time.

Customers are excited at the watch's ability to sync with Outlook's alerts and get continuous content on the move. There have been some teething troubles though: Reiter points out that several users have complained of network outages, with Microsoft's content servers also giving trouble.

Microsoft's official spin on this is that they are working on enhancing the Direct-Band network; the bottom-line is that such wireless networks are going to take a while to reach reliability.

First watches, then the world

Gauging the public reaction to the services, Microsoft looks forward to a point where common everyday objects are going to become smarter and more functional using software. The idea is to have innocuous, intelligent and pervasive devices existing side by side with the desktop. The future might be to have two-way communication.

The moot point is that of acceptability—would you junk your current watch and choose to buy this clunky, possibly more expensive watch? Reiter would tell you that if he'd get a slimmer watch, with a dependable, robust network, and more feeds, he would consider switching over for certain occasions. Until then, his handsome gold and chrome Cartier stays firmly strapped to his wrist. ■

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	1656 Christiaan Huygens invents the pendulum clock
	1914 Wrist watches become commonly adopted as World War I starts
	1952 NIST announces NBS-1, first atomic clock using cesium molecules
	1967 The 13th General Conference on Weights and Measures redefines the second on the vibrations of the cesium atom
	1969 Seiko Astron, the world's first quartz watch debuts
	1973 Introduction of the world's first LCD display watch, a Seiko
	Late 70's Quartz watches and clocks now have alarms, become multifunctional, have calculators built into them
	1980s Watches with calculators become common
	1990s Sports watches start coming with various additional functions
	2000s Watches now move on to become personal organisers
	2004 SPOT debuts commercially